

Week 12: Centralized Logging (Part 2)

Date: May 25, 2026

Study Time: 5-8 hours total

GOAL

Complete centralized logging for all tiers

STUDY TASKS (1-2 hours)

Reading

- ☐ "How Linux Works" by Brian Ward
 - Chapter 7: Log rotation section

Video Learning

- ☐ LearnLinuxTV - Search: "logrotate"
- ☐ Professor Messer - Search: "Log Management"

Concepts to Understand

- ☐ Why rotate logs?
 - ☐ logrotate configuration
 - ☐ Compression of old logs
 - ☐ Log retention policies
 - ☐ grep, awk, sed for log analysis
 - ☐ Log aggregation vs log analysis
 - ☐ Disk space management for logs
-

LAB WORK (3-4 hours)

Forward Logs from ALL Remaining Devices

Last week you set up Pi 4. Now add the rest.

- ☐ Configure Laptop (Ubuntu)

bash

```
# On your laptop:
sudo nano /etc/rsyslog.conf

# Add at end:
*. * @@100.64.0.10:514
# (Use your Fedora's Tailscale IP)

sudo systemctl restart rsyslog

# Test:
logger "TEST FROM LAPTOP"
```

☐ **Configure Windows Desktop (if running WSL)**

bash

```
# In WSL:
sudo nano /etc/rsyslog.conf
# Add same forwarding rule
sudo service rsyslog restart
logger "TEST FROM WINDOWS/WSL"
```

☐ **Configure Raspberry Pi Zero (Tier 3)**

bash

```
ssh pi-zero
sudo nano /etc/rsyslog.conf

# Add:
*. * @@100.64.0.10:514

sudo systemctl restart rsyslog
logger "TEST FROM PI ZERO"
```

☐ **Verify ALL logs arriving on Fedora**

```
bash
```

```
ssh fedora  
sudo tail -f /var/log/messages
```

```
# You should see messages from:  
# - laptop  
# - pi4  
# - pi-zero  
# (and any other devices)
```

Configure Log Rotation

Logs can fill up your disk! logrotate manages this.

☐ Understand current logrotate config

```
bash
```

```
# Main config  
cat /etc/logrotate.conf  
  
# Individual configs  
ls -l /etc/logrotate.d/  
  
# View rsyslog rotation  
cat /etc/logrotate.d/rsyslog
```

☐ Create logrotate config for remote logs

bash

```
sudo nano /etc/logrotate.d/remote-logs

# Add:
/var/log/remote/*/*.log {
    daily
    missingok
    rotate 30
    compress
    delaycompress
    notifempty
    create 0640 root root
    sharedscripts
    postrotate
        systemctl reload rsyslog > /dev/null 2>&1 || true
    endscript
}

# Explanation:
# daily - rotate every day
# missingok - don't error if log missing
# rotate 30 - keep 30 days of logs
# compress - gzip old logs
# delaycompress - don't compress most recent
# notifempty - don't rotate if empty
# create 0640 - permissions for new file
# postrotate - reload rsyslog after rotation
```

☐ Test logrotate manually

bash

```
# Dry run (doesn't actually rotate)
sudo logrotate -d /etc/logrotate.d/remote-logs

# Force rotation (for testing)
sudo logrotate -f /etc/logrotate.conf

# Check it worked
ls -lh /var/log/remote/pi4/
# Should see .log and .log.1.gz files
```

Log Analysis with grep

grep is your best friend for searching logs.

☐ Basic grep patterns

bash

```
# Search for "error" in messages
grep "error" /var/log/messages

# Case-insensitive
grep -i "error" /var/log/messages

# Show line numbers
grep -n "failed" /var/log/messages

# Show 3 lines before and after match
grep -C 3 "error" /var/log/messages

# Count matches
grep -c "ssh" /var/log/messages

# Invert match (show non-matching lines)
grep -v "systemd" /var/log/messages
```

☐ Search across all logs

bash

```
# Search all files in /var/log
grep -r "failed password" /var/log/

# Search only .log files
grep -r --include="*.log" "error" /var/log/
```

☐ Multiple patterns with grep

bash

```
# Match "error" OR "fail"
grep -E "error|fail" /var/log/messages

# Match lines with "ssh" AND "failed"
grep "ssh" /var/log/messages | grep "failed"
```

☐ Practice: Find failed SSH attempts

bash

```
# Ubuntu/Debian:
grep "Failed password" /var/log/auth.log

# Fedora:
grep "Failed password" /var/log/secure

# Count by IP
grep "Failed password" /var/log/secure | awk '{print $(NF-3)}' | sort | uniq -c |
```

Log Analysis with awk

awk is powerful for extracting specific fields.

☐ awk basics

bash

```
# Print specific columns (space-separated)
awk '{print $1, $5}' /var/log/messages
# $1 = first column, $5 = fifth column

# Print last column
awk '{print $NF}' /var/log/messages
# NF = Number of Fields

# Print lines containing "error"
awk '/error/ {print $0}' /var/log/messages
```

☐ Practical awk examples

bash

```
# Extract just timestamps and hostnames
awk '{print $1, $2, $3, $4}' /var/log/messages

# Sum numbers (if logs have metrics)
awk '{sum += $7} END {print sum}' file.log

# Print unique values from column 5
awk '{print $5}' /var/log/messages | sort | uniq
```

☐ Practice: Analyze authentication failures

bash

```
# Who tried to log in and failed?
grep "Failed password" /var/log/secure | awk '{print $9, $11}'
# Field 9 = username, Field 11 = IP

# Count failures by username
grep "Failed password" /var/log/secure | awk '{print $9}' | sort | uniq -c | sort

# Count failures by IP
grep "Failed password" /var/log/secure | awk '{print $11}' | sort | uniq -c | sort
```

Advanced: Combining grep, awk, and sed

☐ Build a log analysis pipeline

bash

```
# Find errors from yesterday, extract time and message
journalctl --since yesterday --until today | \
  grep -i error | \
  awk '{print $1, $2, $3, $NF}' | \
  sort | \
  uniq -c

# Breakdown:
# 1. journalctl gets yesterday's logs
# 2. grep finds lines with "error"
# 3. awk prints timestamp and last field
# 4. sort groups identical lines
# 5. uniq -c counts occurrences
```

Create Log Monitoring Script

☐ Create simple monitoring script

bash

```
nano ~/check-logs.sh

#!/bin/bash
# Simple log monitoring script

LOGFILE="/var/log/messages"
ERRORS=$(grep -c "error" $LOGFILE)
WARNINGS=$(grep -c "warning" $LOGFILE)

echo "=== Log Summary for $(hostname) ==="
echo "Errors: $ERRORS"
echo "Warnings: $WARNINGS"

if [ $ERRORS -gt 10 ]; then
    echo "⚠️ High number of errors detected!"
fi

# Make executable
chmod +x ~/check-logs.sh
./check-logs.sh
```

Verify All Tier Logs on Fedora

☐ Check log directories

bash

```
# On Fedora:
ls -lh /var/log/remote/

# Should have directories for:
# - laptop
# - pi4
# - pi-zero
# - (any other devices)
```

☐ Monitor live logs from all devices

```
bash
```

```
# Watch all incoming logs
sudo tail -f /var/log/messages

# Or watch specific device
sudo tail -f /var/log/remote/pi4/*.log
```

☐ Generate test traffic from each device

```
bash
```

```
# From laptop:
logger -p user.info "Laptop health check OK"

# From pi4:
ssh pi4
logger -p user.info "Pi4 health check OK"

# From pi-zero:
ssh pi-zero
logger -p user.info "Pi-zero health check OK"

# Watch on Fedora - all should appear
```



FLASHCARD COMMANDS

1. `logrotate`
2. `/etc/logrotate.conf`
3. `logrotate -d` (debug/dry-run)
4. `logrotate -f` (force)
5. `grep`
6. `grep -i` (case insensitive)
7. `grep -r` (recursive)
8. `grep -v` (invert)
9. `awk`
10. `awk '{print $1}'`
11. `sed`
12. `tail`
13. `head`
14. `less`
15. `sort`
16. `uniq`
17. `uniq -c`
18. `wc -l` (count lines)

Concepts:

- Why rotate logs?
 - What does compress do in logrotate?
 - What does delaycompress do?
 - How does awk split fields?
 - Difference between grep and awk?
-

EXAM OBJECTIVES

[https://partners.comptia.org/docs/default-source/resources/comptia-linux-xk0-005-exam-objectives-\(3-0\)](https://partners.comptia.org/docs/default-source/resources/comptia-linux-xk0-005-exam-objectives-(3-0))

Relevant: 2.1, 4.1, 4.2, 4.4

QUARTERLY CHECK-IN #2

Message me: "Quiz me on Weeks 7-12"

Topics covered:

- Systemd service management
- Custom systemd services
- Networking fundamentals
- Firewall rules and iptables
- Centralized logging (rsyslog)
- Log rotation and analysis

Self-assessment:

- Confidence level (1-10): _____
 - Topics needing review: _____
 - Ready for scripting? _____
-

NOTES

Devices forwarding to Fedora:

- ☐ Ubuntu Laptop
- ☐ Windows Desktop (WSL)
- ☐ Raspberry Pi 4
- ☐ Raspberry Pi Zero
- ☐ Other: _____

Log rotation schedule:

- Rotation frequency: _____ (daily/weekly)
- Days to keep: _____
- Compression enabled? _____

Disk space for logs:

```
bash
```

```
# Check current usage
```

```
df -h /var/log
```

```
du -sh /var/log/remote/
```

Current usage: _____

grep/awk practice results: (What patterns did you find most useful?)

Completed: ☐

QUICK REFERENCE

logrotate Config Options:

```
daily/weekly/monthly - Rotation frequency
rotate N - Keep N old logs
compress - Gzip old logs
delaycompress - Don't compress most recent
notifempty - Don't rotate empty files
missingok - Don't error on missing files
create MODE OWNER GROUP - Permissions for new file
postrotate...endscript - Commands after rotation
```

grep Cheat Sheet:

```
bash
```

```
grep pattern file           # Basic search
grep -i pattern file        # Case insensitive
grep -r pattern dir/        # Recursive
grep -v pattern file        # Invert (exclude)
grep -n pattern file        # Show line numbers
grep -c pattern file        # Count matches
grep -l pattern files       # List files with matches
grep -A 3 pattern file      # 3 lines after
grep -B 3 pattern file      # 3 lines before
grep -C 3 pattern file      # 3 lines before/after
grep -E "p1|p2" file        # Multiple patterns (OR)
```

awk Basics:

bash

```
awk '{print $1}' file           # First field
awk '{print $NF}' file          # Last field
awk '/pattern/ {print}' file    # Lines matching pattern
awk '{sum+=$3} END {print sum}' file # Sum column 3
awk -F: '{print $1}' /etc/passwd # Use : as delimiter
```

Common Log Locations:

- /var/log/messages - General system
- /var/log/syslog - System messages (Ubuntu)
- /var/log/auth.log - Authentication (Ubuntu)
- /var/log/secure - Authentication (Fedora)
- /var/log/kern.log - Kernel
- /var/log/cron - Cron jobs
- /var/log/remote/ - Your centralized logs

Previous Week: Centralized Logging Setup

Next Week: Bash Scripting Basics